Variational Autoencoders

We know the process of a Autoencoder

Given an x it generates a latent space representation of the input x as z

So this representation z is capable of giving much information about the input

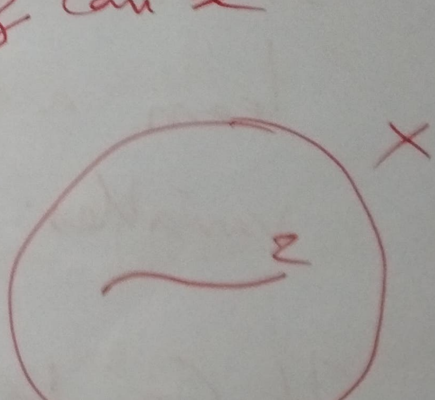
but the problem is that both the ~~generator and~~ encoder and decoder are deterministic in nature

We provide an input x it generates a hidden representation z, that to a unique one. Similarly the decoder if you input a latent representation z it will output the 'x' back

Now the question is whether we can use this autoencoder as a generative model

The idea is that the hidden representations are infact lying on a subspace of the input space

Now Assume we are interested in image generation. So to generate similar images as that you have used to train your autoencoder. using the decoder part what hidden representation z can be used?



z a subspace of a higher dimensional space x (input)

So we can use those z that are highly likely to output x's

that means we are interested in sampling from $P(z/x)$ so that we choose those z that have high probability.

So directly Autoencoder cannot fit like a generative model

So VAE variational Autoencoder are Autoencoders which have the same structure as a autoencoder but they learn a distribution over the hidden variables.

It can be thought of a probabilistic version of autoencoder

Let $\{X = x_i\}_{i=1}^{N}$ be the training data

then X can be thought of as a random

variable in R^n.

hidden

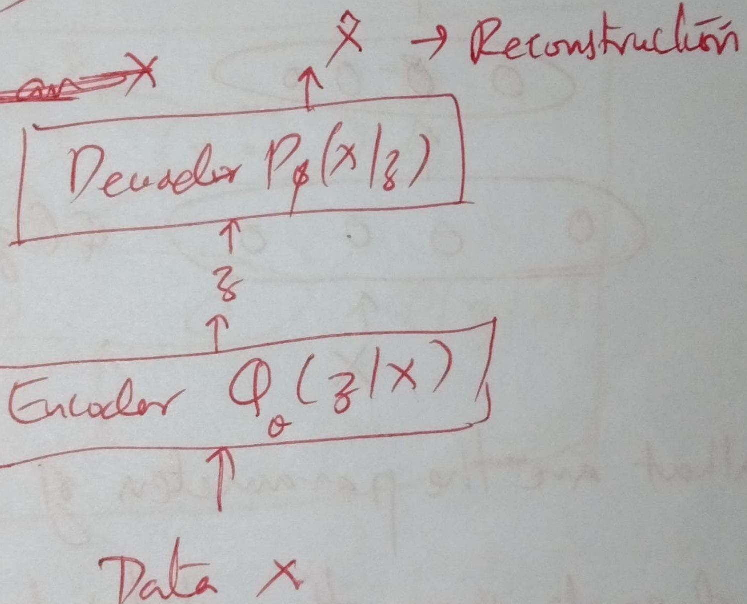
visible variable

Here we are interested to learn $P(z|x)$

and $P(x|z)$,

~~ie given an~~ ~~x~~

ie,



θ - parameters of a encoder network

ϕ - parameters of decoder network.

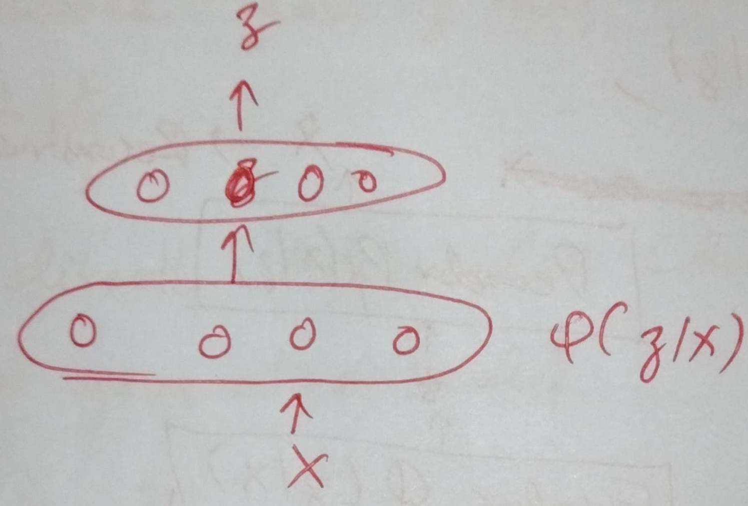
ie, learn a distribution over latent variables

$q(z|x)$ (VAE uses a neural network to do that)

Second $P(x|z)$ (VAE uses a neural network to do that)

Encoder: what do you mean when we say that we want to learn a distribution

Ans: We mean to learn the parameters of the distribution



What are the parameters of $\varphi(z/x)$

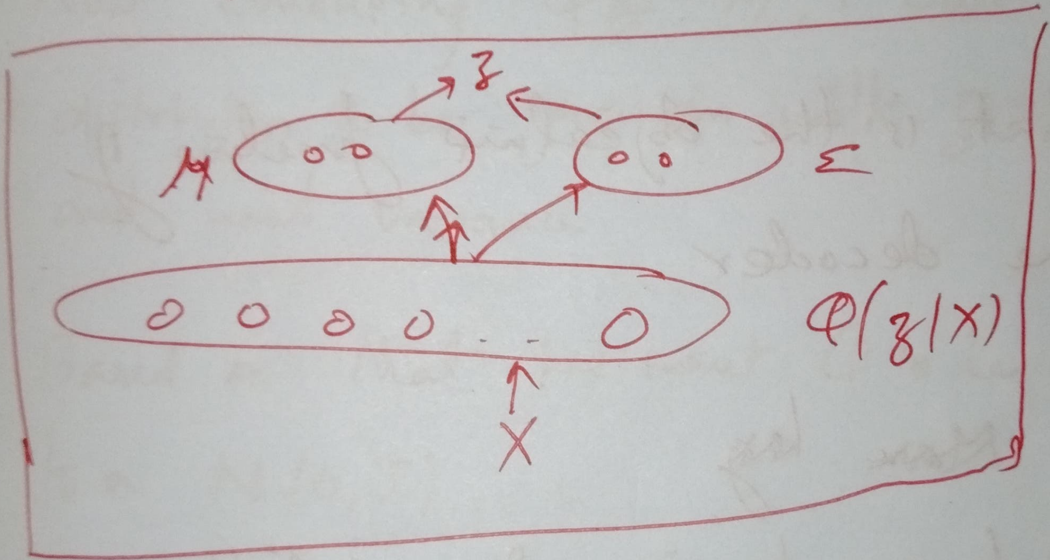
it depends upon the distribution we are ~~assuming~~ assuming.

In VAE we assume that latent variable come from a standard normal distribution $N(0, I)$ and job of the encoder is to predict the parameters of this distribution

Given a x the job of encoder is to give output as the parameters μ and Σ.

$$\mu, \Sigma = f_\theta(x)$$

Once i know what is μ, and Σ then i can sample z from the same distribution



Now what is the input to the decoder

We have to sample from the distribution and now the decoder is supposed to learn $P(X|z)$.

Here again we assume that its a gaussian distribution with unit variance ie $N(\mu, \Sigma)$ with $\Sigma = I$

Then ~~we~~ need to predict the mean of the distribution

$$\therefore \mu = f_\phi(z)$$

$\hookrightarrow$ decoder

$\therefore$ what is the objective function of the decoder

~~Max~~ log

w.r.t to one training sample x_i, the objective function is Minimize

$$- E_{z \sim q_\theta(z|x_i)} \log p_\phi(x_i|z)$$

$\hookrightarrow$ This is nothing but, the decoder tries to reconstruct the original data sample x_i from a latent space representation z.

So ~~the~~ we want the decoder to

assign high probability to the true data x_i

$\therefore$ The total loss in the Sum of all the ~~data~~ cases ~~or~~ i,e, over all the data points

Now Considering $Q_\theta(z|x)$, i want the distribution to be gaussian with zero mean and unit variance.

Based on that we want Q to be as close to a $N(0, I)$

We have the notion of KL divergence between two distribution. loss function is to min

$$\therefore l_i = -E_{z \sim Q_\theta(z|x_i)}[\log P_\phi(x_i|z)] + KL(Q_\theta(z|x_i) \| P(z))$$

Distribution P: Person 1 believes a fair coin has an prob of 80% of landing on heads P(Heads = 0.8) and a 20% prob of landing on tails P(Tails) = 0.2

Distribution Q: Person 1 P(heads = 0.7)

KL Divergence $D_{KL}(P \| Q)$

$$= P(\text{Heads}) \cdot \log \frac{P(\text{Heads})}{Q(\text{Heads})}$$

$$+ P(\text{Tails}) \cdot \log \frac{P(\text{Tails})}{Q(\text{Heads})}$$

This represents the Information loss when when using Person 2's beliefs (Q) to represent person 1's belief (P)

$$\text{Here value} = 0.8 \cdot \log \frac{0.8}{0.7} + 0.2 \log \frac{(0.2)}{0.3}$$

$$\approx \underline{\underline{0.0257}}$$

If distributions are identical KL divergence value is 0.

$$\boxed{D_{KL}(P \| Q) = \int P(x) \log \frac{P(x)}{Q(x)} \, dx}$$

Continious Case

$$\boxed{D_{KL}(P \| Q) = \sum_{x} P(x) \log \frac{P(x)}{Q(x)}}$$

Discrete Case